

0.1 Logic and Quantifiers

Mathematical statements in linear algebra often involve claims about “all” vectors or the “existence” of a solution. For instance, saying a matrix equation $\mathbf{Ax} = \mathbf{b}$ has a solution means *there exists* a vector $\mathbf{x}$ satisfying the equation; saying a set is linearly independent means *for all* nontrivial linear combinations, the result is nonzero. To work with these statements precisely—especially when proving or disproving them—we need the language of **quantifiers**.

0.1.1 Quantifiers

Definition 0.1.1 — Quantifiers

Let $P(x)$ be a statement (predicate) that depends on a variable x ranging over some set S .

- The **universal quantifier** $\forall$ means “for all” or “for every.” The statement

$$\forall x \in S : P(x)$$

asserts that $P(x)$ is true for *every* element x in S .

- The **existential quantifier** $\exists$ means “there exists” or “for some.” The statement

$$\exists x \in S : P(x)$$

asserts that $P(x)$ is true for *at least one* element x in S .

Remark. We sometimes write $\forall x \in S, P(x)$ or $(\forall x \in S) P(x)$ instead of $\forall x \in S : P(x)$. The notation $\exists! x \in S : P(x)$ means “there exists a **unique** x in S such that $P(x)$.”

Example. (Quantified Statements)

Consider the following statements:

1. $\forall x \in \mathbb{R} : x^2 \geq 0$.
This says “every real number squared is nonnegative.” This is **true**.
2. $\exists x \in \mathbb{R} : x^2 = -1$.
This says “there exists a real number whose square is -1 .” This is **false** in $\mathbb{R}$, but becomes true if we replace $\mathbb{R}$ with $\mathbb{C}$.
3. $\forall x \in \mathbb{R} : x < x + 1$.
This is **true** for all real numbers.
4. $\exists n \in \mathbb{Z} : n^2 = 2$.
This is **false**—there is no integer whose square is 2.

Quantifiers can be **nested**: statements may involve multiple quantifiers in sequence. The **order matters**.

Example. (Nested Quantifiers)

Consider these two statements about real numbers:

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : y > x$.
 “For every real number x , there exists a real number y greater than x .” This is **true**—given any x , we can take $y = x + 1$.
2. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R} : y > x$.
 “There exists a real number y that is greater than all real numbers.” This is **false**—no such largest real number exists.

The order of quantifiers changes the meaning entirely.

0.1.2 Negating Quantified Statements

The key to proof by contradiction is knowing how to **negate** a statement. When negating quantified statements, there is a systematic rule: the universal quantifier becomes existential, and vice versa, while the predicate gets negated.

Definition 0.1.2 — Negation of Quantifiers

The negation of a quantified statement follows these rules:

$$\neg(\forall x \in S : P(x)) \equiv \exists x \in S : \neg P(x)$$

$$\neg(\exists x \in S : P(x)) \equiv \forall x \in S : \neg P(x)$$

In words:

- “Not all x satisfy P ” is equivalent to “there exists an x that does not satisfy P .”
- “There does not exist an x satisfying P ” is equivalent to “all x fail to satisfy P .”

Example. (Basic Negation Examples)

Here are two simple examples of negating quantified statements:

1. The negation of “ $\forall x \in \mathbb{R} : x^2 \geq 0$ ” is “ $\exists x \in \mathbb{R} : x^2 < 0$.”
 The original is true; the negation is false.
2. The negation of “ $\exists n \in \mathbb{N} : n$ is even” is “ $\forall n \in \mathbb{N} : n$ is odd.”
 The original is true (take $n = 2$); the negation is false.

The following examples demonstrate how to negate definitions that you will encounter repeatedly in linear algebra.

Example. (Negating the Definition of Injectivity)

Recall that a function $f : A \rightarrow B$ is **injective** if:

$$\forall a_1, a_2 \in A : f(a_1) = f(a_2) \Rightarrow a_1 = a_2$$

To prove f is **not** injective, we negate this statement. Since $P \Rightarrow Q$ is equivalent to $\neg P \vee Q$, its negation is $P \wedge \neg Q$. Thus:

$$\exists a_1, a_2 \in A : f(a_1) = f(a_2) \wedge a_1 \neq a_2$$

To show a function is not injective, find two **distinct** elements that map to the same value.

Example. (Negating the Definition of Surjectivity)

A function $f : A \rightarrow B$ is **surjective** if:

$$\forall b \in B, \exists a \in A : f(a) = b$$

The negation is:

$$\exists b \in B, \forall a \in A : f(a) \neq b$$

To prove f is **not** surjective, find a single element in the codomain that no element of the domain maps to.

Remark. Mastering quantifier negation is essential for proof-based mathematics. Many proofs in linear algebra fall into one of the following patterns:

- Showing something holds for “all” vectors: use a direct proof starting with “let v be an arbitrary vector...”
- Showing something “exists”: construct it explicitly, or use contradiction by assuming nothing exists.
- Disproving a “for all” statement: find a single counterexample.
- Disproving an “exists” statement: show that every candidate fails.